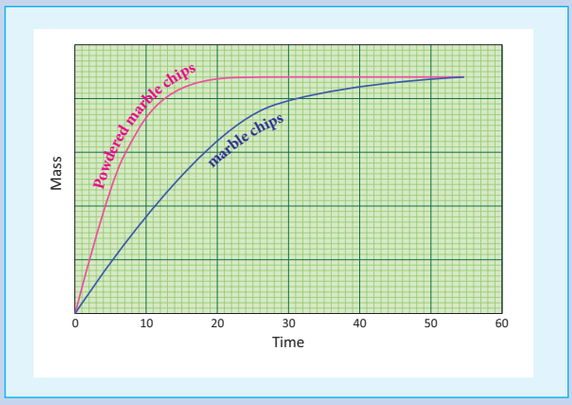
## Activity

A Known mass of marble chips are taken in a flask and a known volume of dilute HCl is added to the content, a stop clock is started when half the volume of HCl is added. The mass is noted at regular intervals until the reaction is complete. Same experiment is repeated with the same mass of powdered marble chips and the observations are recorded.

Reaction

$$CaCO_3(s) + 2HCl\,(aq) \longrightarrow CaCl_2(aq) + H_2O\,(l) + CO_2(g)$$

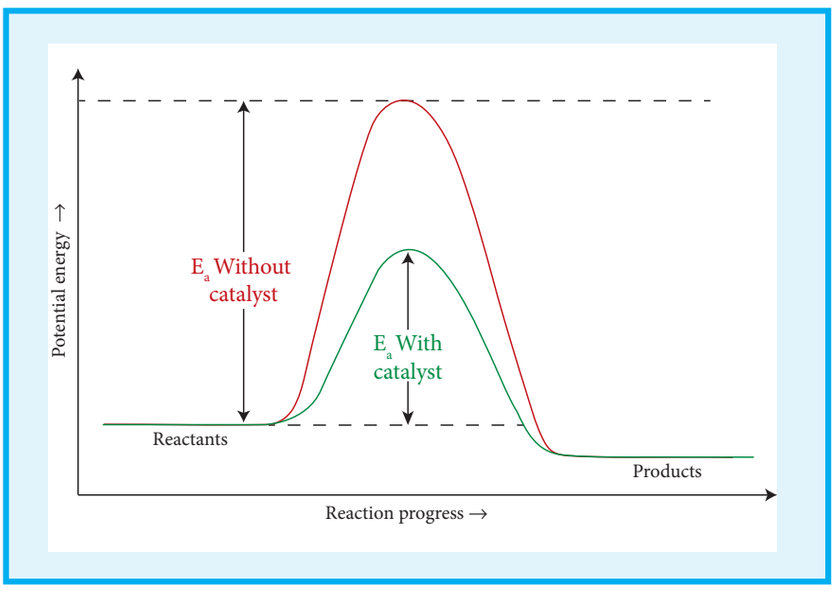
Since, carbon dioxide escapes during reaction, the mass of the flask gets lighter as the reaction proceeds. So, by measuring the flask, we can follow the rate of the reaction. A plot of loss in mass Vs time is drawn and it looks like the one as shown below.




For the powdered marble chips, the reaction is completed in less time. i.e., rate of a reaction increases when the surface area of a solid reactant is increased.

### 7.9.4 Effect of presence of catalyst:

So far we have learnt, that rate of reaction can be increased to certain extent by increasing the concentration, temperature and surface area of the reactant. However significant changes in the reaction can be brought out by the addition of a substance called catalyst. A catalyst is substance which alters the rate of a reaction without itself undergoing any permanent chemical change. They may participate in the reaction, but again regenerated and the end of the reaction. In the presence of a catalyst, the energy of activation is lowered and hence, greater number of molecules can cross the energy barrier and change over to products, thereby increasing the rate of the reaction.






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